

Simultaneous Multi-Rate Threshold Optimization of Protograph-Based Raptor-Like LDPC Codes via Genetic Algorithm

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Abstract—Protograph-based Raptor-like (PBRL) LDPC codes, adopted in the 5G NR eMBB data channel, support a wide range of code rates by generating incremental redundancy through XOR operations. As the protograph structure directly influences performance, its efficient design is crucial. This letter introduces a novel framework for optimizing the IRC submatrix by minimizing the Euclidean norm of capacity gaps, a new evaluation metric that captures overall threshold behavior across all supported rates, in contrast to the conventional rate-by-rate greedy optimization. The approach formulates this as a global optimization problem solved via a genetic algorithm (GA) with simulated annealing (SA) applied for improved convergence. Numerical results show notable threshold reductions, including a 58.4% improvement in the capacity gap norm for the 5G Base Graph 1. The method is also compatible with other rate-compatible structures, such as rate-compatible spatially coupled LDPC codes, which are promising candidates for coding schemes in next-generation mobile systems.

Index Terms—PBRL LDPC codes, protograph optimization, decoding threshold, genetic algorithm, 5G NR.

I. INTRODUCTION

HYBRID ARQ with incremental redundancy (IR-HARQ) protocols plays a crucial role in modern wireless communication systems such as LTE and 5G NR, relying on rate-compatible (RC) codes to support multiple code rates using a single encoder and decoder structure [6].

A well-known class of RC codes is the protograph-based Raptor-like (PBRL) LDPC codes proposed by Chen et al. [1], which outperform standardized RC LDPC codes such as AR4JA and DVB-S2 at comparable block lengths. A PBRL protograph is composed of a highest-rate code (HRC) and an incremental redundancy code (IRC). The HRC provides a reliable structure for the highest-rate code in the PBRL family, while the IRC generates additional parity bits, thereby lowering the code rate. The conventional IRC design strategy optimizes the connections of each supported rate sequentially, determining the non-zero entries of one row in the IRC sub-matrix at a time, starting from the highest rate and progressively adding rows to minimize the decoding threshold at the current rate. Additionally, several studies have applied heuristic and evolutionary algorithms to various aspects of protograph-based LDPC design, such as threshold

minimization [7], degree distribution optimization and short-cycle minimization for rate-adaptive source coding [10], protograph optimization for improved waterfall and low error floors [11], or direct error-rate minimization for short block lengths [12].

While heuristic algorithms have been successfully applied to LDPC design, they have focused on single-rate design or non-PBRL structures. Furthermore, the sequential optimization in conventional PBRL protograph design often leads to increased gaps to capacity at lower rates, due to limited degrees of freedom and the inability to refine prior rows of the IRC sub-matrix once they are fixed. This results in suboptimal and unbalanced threshold performance across supported rates, as early decisions cannot be changed. To the best of our knowledge, no prior work has formulated the optimization of the IRC component as a global multi-rate problem.

To overcome this limitation, we propose a joint multi-rate optimization framework that selects the elements of all rows of the IRC sub-matrix simultaneously. To solve this discrete problem, we employ a genetic algorithm (GA) [8] to explore the solution space effectively. Within the fitness function, the decoding threshold at each rate is computed using Protograph-EXIT (PEXIT) [9]. To further improve convergence and avoid local minima, we apply simulated annealing (SA) initialized with the best GA solution.

Throughout this letter, the terms protograph (base graph) and protomatrix (base matrix) are used interchangeably where the protomatrix is the matrix representation of the protograph (rows as check nodes and columns as variable nodes). The rest of the letter is organized as follows. Sec. II reviews the PBRL structure and the design of HRC and IRC components. Sec. III presents the proposed optimization method, including the formulation, GA and SA algorithms. Sec. IV provides numerical evaluations on various protographs from the literature. Finally, Sec. V concludes the letter.

II. PBRL CODE STRUCTURE AND DESIGN PRINCIPLES

A. PBRL Code Structure

The PBRL code structure combines an HRC protograph and an IRC protograph into a single bipartite protograph. The protomatrix structure of the PBRL code

$$B = \begin{bmatrix} B_{\text{HRC}} & 0 \\ B_{\text{IRC}} & I \end{bmatrix} \quad (1)$$

consists of four sub-matrices: B_{HRC} , defining a small high-rate protograph; B_{IRC} , enabling rate compatibility via degree-one variable nodes; 0 and I , which are zero and identity matrices of appropriate sizes, determined by the number of supported

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rates, respectively. Puncturing is a rate-adaptation technique that omits specified bits from transmission, yielding a shorter codeword without changing the encoder and decoder structures [2]. At the receiver, punctured bits are initialized to a fixed LLR, and the iterative decoding attempts to recover them via parity constraints. In PBRL codes, a variant known as punctured-node PBRL (PN-PBRL) punctures one or more variable nodes in the HRC submatrix, effectively lowering the design rate without changing the base graph topology. This puncturing of HRC nodes has been shown to improve iterative decoding thresholds by shifting the waterfall region closer to the Shannon limit [3]. Throughout the letter, PBRL refers to the PN-PBRL family unless stated otherwise. A PBRL code family is specified by the triplet (n_v, n_c, n_p) where n_v , n_c , and n_p denote the numbers of variable nodes, check nodes, and punctured nodes in the HRC part, respectively. Rate-compatibility is achieved by appending successive IRC check nodes and their associated degree-one variable nodes to a fixed high-rate protograph. Transmitting these additional nodes one at a time produces a family of discrete code rates:

$$R_i = \frac{n_v - n_c}{n_v - n_p + i}, i = 1, \dots, n_r, \quad (2)$$

where i denotes the number of activated rows in the IRC submatrix. Extensive studies have shown that, when each IRC row is optimally connected to the HRC, the resulting PBRL ensemble achieves near-capacity thresholds uniformly across all rates without structural modifications.

B. PBRL Protograph Design

As shown in (1), designing a PBRL protograph reduces to determining its two sub-matrices, B_{HRC} and B_{IRC} . The highest-rate code B_{HRC} is constructed as follows. First, a triplet (n_v, n_c, n_p) is chosen. Following [4], all variable-node degrees are constrained to be at least three to ensure linear minimum-distance growth after lifting. In [1], the authors focused on varying the number of degree-4 variable nodes, as nodes of degree 4 offer a good balance between threshold performance and error-floor behavior. Finally, a modified Reciprocal Channel Approximation (RCA) is used to evaluate candidate B_{HRC} protographs and select the one that has the lowest decoding threshold. For instance, in the example with $k = 1032$ information bits in [1], the protograph with seven degree-4 nodes provided the best trade-off and was chosen as the HRC base graph.

After fixing B_{HRC} , each row of B_{IRC} associated with a new degree-one variable node is appended one by one to the HRC part. Consequently, each IRC row is treated as a separate optimization problem for the specification of its non-zero entries, with the ultimate goal of selecting the connections between the IRC and HRC parts so as to minimize the iterative decoding threshold at the corresponding rate. The design enforces constraints to ensure linear minimum-distance growth after lifting, to control harmful graph structures. Typical constraints, adopted in prior designs [1], [4], are:

- 1) The minimum degree of each added check node (an IRC row) is at least two.

- 2) The punctured variable node must be connected to all (or almost all) IRC rows.
- 3) Parallel connections to the punctured variable node are employed to improve the threshold performance.

However, constraints may vary slightly depending on the target scenario. Let n_r denote the number of supported code rates, and let B_{IRC} consist of rows $b_1, \dots, b_{n_r}$. We denote by $T(b_1, \dots, b_i)$ the decoding threshold at the i -th rate R_i , evaluated on the full PBRL protomatrix composed of the fixed B_{HRC} , the first i rows of B_{IRC} , and their associated degree-one columns. In the sequential design of Chen et al. [1], for $i = 1, \dots, n_r$, each IRC row is selected one at a time according to

$$b_i^* = \arg \min_{b_i \in \mathcal{H}^{n_v}} T(b_1^*, \dots, b_{i-1}^*, b_i) \quad \text{s.t. } b_i \text{ satisfies constraints 1 to 3,} \quad (3)$$

where prior rows $b_1^*, \dots, b_{i-1}^*$ have been obtained already and $\mathcal{H}$ denotes the allowable set of elements (e.g., $\{0, 1\}$ or $\{0, 1, 2\}$). The complete IRC submatrix is then

$$B_{\text{IRC}}^* = [b_1^*, \dots, b_{n_r}^*]^\top. \quad (4)$$

III. PROPOSED JOINT IRC PART OPTIMIZATION

While the conventional greedy approach ensures local optimality at each code rate, it cannot adapt earlier row decisions to later additions. To address this limitation, we propose a joint multi-rate threshold optimization framework that simultaneously specifies all entries of B_{IRC} in the PBRL protograph. Unlike prior methods [1], [5] that sequentially minimize the threshold for each row while fixing previous ones, our unified single-stage formulation employs a new objective: minimizing the Euclidean norm of the gap-to-capacity vector across all supported rates. This yields a more balanced, near-capacity threshold profile over the rate family. Let R_i denote the code rate after appending i rows to B_{IRC} , and let $S(R_i)$ be the corresponding Shannon limit. The gap vector is defined as:

$$\mathbf{g}(b_1, \dots, b_{n_r}) = [T(b_1) - S(R_1), T(b_1, b_2) - S(R_2), \dots, T(b_1, \dots, b_{n_r}) - S(R_{n_r})]^\top. \quad (5)$$

The joint optimization problem is formulated as:

$$\{b_i^*\}_{i=1}^{n_r} = \arg \min_{b_1, \dots, b_{n_r} \in \mathcal{H}^{n_v}} \|\mathbf{g}(b_1, \dots, b_{n_r})\|_2 \quad \text{s.t. each constraints 1 to 3 from Sec. II.} \quad (6)$$

We employ a genetic algorithm to solve the joint optimization problem, which outperforms the prior greedy method. To avoid the high cost of density evolution on AWGN channels, we employ the fast and accurate PEXIT method. Our focus is solely on determining the rows of B_{IRC} while B_{HRC} , the identity block and zero block remain fixed. We use the Euclidean norm $\|\cdot\|_2$, which has been widely adopted in multi-objective optimization to produce well-balanced solutions. Since our decision variables b_i are vectors from a discrete space and the objective relies on PEXIT computations, gradient-based methods like gradient descent are inapplicable. Genetic algorithm (GA) is highly effective for exploring large discrete spaces

Algorithm 1 GA-Based Joint IRC Optimization**Require:** $B_{\text{HRC}}, n_r, n_v, P, p_c, p_m, G_{\text{max}}$ **Ensure:** B_{IRC}^*

- 1: Encode each $B_{\text{IRC}} \in \mathcal{H}^{n_r \times n_v}$ as chromosome $\mathbf{c}$
- 2: Initialize population $\{\mathbf{c}_j\}_{j=1}^P$ by sampling each gene $\in \mathcal{H}$
- 3: **for** $g = 1$ to G_{max} **do**
- 4: **for** each chromosome $\mathbf{c}_j$ **do**
- 5: Reshape $\mathbf{c}_j \rightarrow B_{\text{IRC}}$
- 6: Compute $f_j \leftarrow \|g(b_1, \dots, b_{n_r})\|_2$
- 7: **end for**
- 8: Select parents via roulette-wheel selection using probabilities $p_j = \frac{\exp(-\beta f_j / f_{\text{max}})}{\sum_{i=1}^P \exp(-\beta f_i / f_{\text{max}})}$.
- 9: Apply crossover(p_c) and mutation(p_m) to produce offspring
- 10: Apply elitism: retain top P individuals from the combined parent and offspring populations.
- 11: **end for**
- 12: **return** best chromosome reshaped to B_{IRC}^*

without derivative information. In a GA, a chromosome represents a candidate solution to the optimization problem, typically encoded as a fixed-length vector. Each entry in this vector is called a gene and corresponds to a single entry in B_{IRC} , representing the number of edges between a particular variable and check node in the protograph. We encode each candidate $B_{\text{IRC}} \in \mathcal{H}^{n_r \times n_v}$ as a chromosome

$$\mathbf{c} = [b_{1,1}, \dots, b_{1,n_v}, \dots, b_{n_r,1}, \dots, b_{n_r,n_v}]. \quad (7)$$

An initial population of P chromosomes is sampled uniformly from $\mathcal{H}$. Fitness is evaluated by reshaping $\mathbf{c}$ into B_{IRC} and computing the norm of the gap vector $\|g(b_1, \dots, b_{n_r})\|_2$. Each generation proceeds as follows:

- 1) **Selection:** The roulette-wheel selection mechanism operates by assigning selection probabilities to individual chromosomes (see Step 8 in Alg. 1), balancing solution exploitation and maintaining population diversity. In this mechanism β is a selection control parameter.
- 2) **Crossover:** p_c is the percentage of selected parents from population. For each selected parent pair, either single-point or two-point crossover is performed by randomly choosing one or two cut points across the chromosome vector and swapping segments to produce two offspring.
- 3) **Mutation:** With probability p_m , each gene is replaced by a random element from $\mathcal{H}$, which helps the population escape local optima and explore novel protomatrix structures.
- 4) **Elitism:** ensures the top P fittest individuals are carried unchanged into the next generation, preserving the best solutions.

The algorithm terminates when a maximum generation count G_{max} is reached. A summary of the proposed joint IRC optimization is provided in Algorithm 1.

To improve convergence and avoid local minima, we apply a simulated annealing (SA) post-processing step to the best

TABLE I
NORM OF GAPS ACROSS DIFFERENT PROTOGRAPH INSTANCES

Instance	Baseline Norm	Proposed Norm	Improvement
(8, 2, 0)	3.6641	1.7451	52.4 %
(10, 2, 1)	1.8159	1.3918	23.4 %
(11, 3, 1)	0.7721	0.7246	6.2 %
5G BG1	1.9210	0.7995	58.4 %
RL-RC-SC	3.1226	2.7967	10.43 %

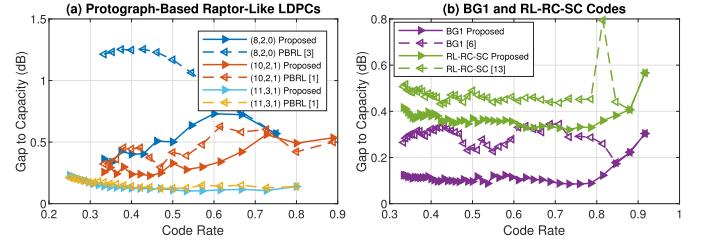


Fig. 1. Gap-to-capacity results for various protographs, including 5G BG1 and an RL-RC-SC LDPC code.

chromosome returned by the GA. Following Thorpe [7], SA perturbs one or more entries in B_{IRC} to explore neighboring solutions and accepts a new candidate B'_{IRC} with probability

$$P_{\text{acc}} = \exp\left(-\frac{\|g(B'_{\text{IRC}})\|_2 - \|g(B_{\text{IRC}})\|_2}{\tau_i}\right), \quad (8)$$

where $\|g(\cdot)\|_2$ is the Euclidean norm of the capacity-gap vector computed for the full protograph (fixed B_{HRC} and the given B_{IRC}), and τ_i is the temperature at iteration i , updated via $\tau_i = \alpha \tau_{i-1}$. The algorithm terminates when the maximum number of iterations is reached. In tested cases, SA provided consistent refinements over the GA output.

GA parameters were set to $P = 50$, $p_c = 0.8$, $p_m = 0.02$, $\beta = 8$ and, $G_{\text{max}} = 200$. SA was applied with $\alpha = 0.95$, $\tau_0 = 10000$ and $I = 5000$ iterations.

The computational complexity is dominated by PEXIT threshold evaluation, which depends on the protograph size. The GA requires $P \times G_{\text{max}} \times n_r$ threshold evaluations, while SA needs $I \times n_r$. Exhaustive search over a space of size $q^{n_r \times n_v}$ is exponentially complex and infeasible for moderate sizes. We implemented parallelized threshold evaluations on an 8-core computer, enabling simultaneous computation of thresholds for up to 8 rates. This resulted in approximately $8 \times$ speedup compared to the serial evaluation. Such parallelization makes the method computationally feasible for protographs of moderate size typically encountered in practice. However, the method can become computationally heavy for large protographs with a high number of variable nodes n_v . In these cases, access to additional computing cores or faster threshold estimation techniques would be necessary. Moreover, while the Euclidean norm used as the objective function is computationally efficient, higher-dimensional objective spaces may mildly increase complexity. Further improvements using alternative objective formulations or enhanced search strategies could help reduce complexity in such scenarios.

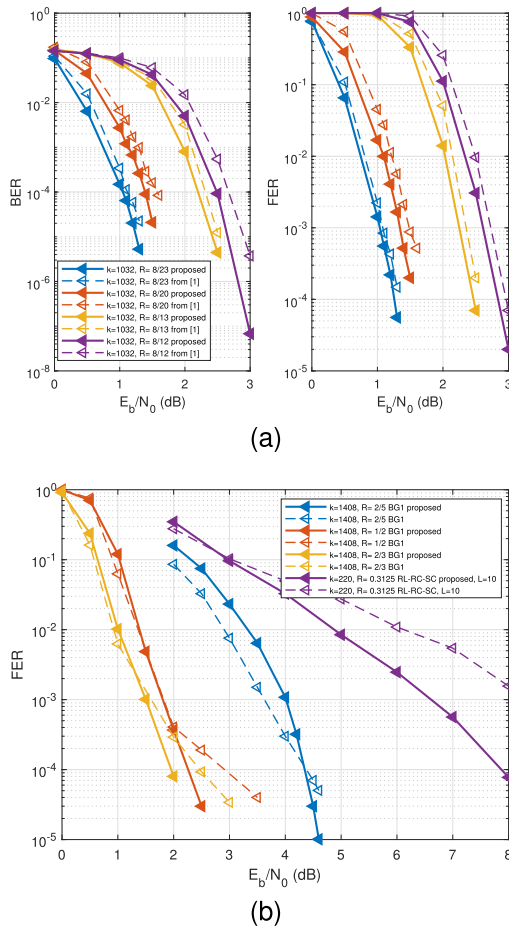


Fig. 2. (a) BER/FER comparison of lifted baseline and proposed (10, 2, 1) PBRL protographs for $k = 1032$. (b) FER performance of lifted BG1 and proposed BG1 ($k = 1408$) and baseline and proposed RL-RC-SC LDPC codes with $k = 220$.

IV. NUMERICAL RESULTS AND PERFORMANCE EVALUATION

We evaluated the proposed method from Sec. III on a diverse set of PBRL protographs from the literature, including designs with 0, 1, or 2 punctured nodes and varying information block lengths k . The benchmark protographs evaluated against their optimized versions, are

- (8, 2, 0) PBRL protograph with $k = 192$ from [3],
- (10, 2, 1) PBRL protograph with $k = 1032$ from [1], Sec. IV-A,
- (11, 3, 1) PBRL protograph with $k = 16384$ from [1], Sec. IV-B,
- The 5G NR Base Graph 1 (BG1), a (26, 4, 2) PBRL protograph with varying information block lengths, as defined in [6].

As an additional application, we applied the optimized BG1 into the Raptor-Like Rate-Compatible Spatially Coupled LDPC (RL-RC-SC-LDPC) structure from [13], using a coupling chain length of $L = 10$ and a coupling width of $m_s = 1$. To isolate the effect of IRC optimization, we fixed the B_{HRC} in each instance. As shown in Table I, our method achieves substantial reductions in the Euclidean norm of capacity gaps across all cases, including a 58.4% improvement for 5G BG1. By contrast, the (11, 3, 1) instance

shows only a 6.2% reduction because the conventional design already achieves near-capacity thresholds by allowing parallel edges in non-punctured nodes for long block-length settings. The decoding thresholds are shown in detail in Fig. 1, illustrating that reducing the Euclidean norm of the capacity gaps results in consistent threshold improvements across nearly all supported code rates. Both baseline and proposed (10, 2, 1) protographs were lifted with degree 129 and information block length $k = 1032$; FER and BER performances over various rates are shown in Fig. 2a. Baseline BG1 and proposed BG1 were lifted with degree 64 and information block length $k = 1408$; FER performances over various rates are in Fig. 2b. For a fair comparison, all protograph instances were lifted using the same method. The results demonstrate that, under comparable protograph constraints, a parity-check matrix constructed using an effective lifting method achieves superior FER and BER performance.

V. CONCLUSION

We introduced a new optimization framework for the IRC part of PBRL protographs by defining a new objective function and applying GA and SA algorithms. Evaluations on standard protographs, including 5G BG1, show significant decoding threshold improvements. The optimized BG1 also yields better thresholds in spatially coupled structures. Finite-length BER and FER simulations demonstrate that the proposed threshold reductions yield superior performance across multiple rates when an effective lifting method is applied.

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